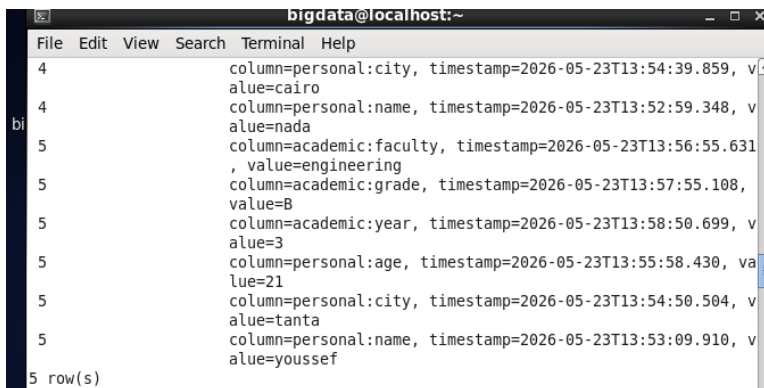


```
hbase(main):035:0> scan 'student'
ROW          COLUMN+CELL
1            column=academic:faculty, timestamp=2026-05-23T13:56:14.238
, value=engineering
1            column=academic:grade, timestamp=2026-05-23T13:57:06.087,
value=A
1            column=academic:year, timestamp=2026-05-23T13:58:06.547, v
alue=3
1            column=personal:age, timestamp=2026-05-23T13:55:02.007, va
lue=21
1            column=personal:city, timestamp=2026-05-23T13:53:47.328, v
alue=cairo
1            column=personal:name, timestamp=2026-05-23T13:50:10.879, v
alue=ali
2            column=academic:faculty, timestamp=2026-05-23T13:56:25.321
, value=medicine
2            column=academic:grade, timestamp=2026-05-23T13:57:22.832,
value=B
2            column=academic:year, timestamp=2026-05-23T13:58:17.493, v
alue=4
2            column=personal:age, timestamp=2026-05-23T13:55:16.665, va
lue=22
2            column=personal:city, timestamp=2026-05-23T13:53:57.333, v
alue=alex
```

Q1. How many rows are in the student table?

Hint: There is a command that counts rows directly.

Answer: 5 Rows



```
bigdata@localhost:~
File Edit View Search Terminal Help
4            column=personal:city, timestamp=2026-05-23T13:54:39.859, v
alue=cairo
4            column=personal:name, timestamp=2026-05-23T13:52:59.348, v
alue=nada
5            column=academic:faculty, timestamp=2026-05-23T13:56:55.631
, value=engineering
5            column=academic:grade, timestamp=2026-05-23T13:57:55.108,
value=B
5            column=academic:year, timestamp=2026-05-23T13:58:50.699, v
alue=3
5            column=personal:age, timestamp=2026-05-23T13:55:58.430, va
lue=21
5            column=personal:city, timestamp=2026-05-23T13:54:50.504, v
alue=tanta
5            column=personal:name, timestamp=2026-05-23T13:53:09.910, v
alue=youssef
5 row(s)
```

Q2. Get only the personal:city column for student with row key 3.

```
hbase(main):036:0> get 'student', '3', {COLUMN => 'personal:city'}
COLUMN      CELL
personal:city timestamp=2026-05-23T13:54:26.754, value=giza
1 row(s)
Took 0.0219 seconds
```

Q3. Scan the table and return only academic:faculty, starting from row 2, with a limit of 3 rows.

```
hbase(main):037:0> scan 'student', {
hbase(main):038:1*   COLUMNS => 'academic:faculty',
hbase(main):039:1*   STARTROW => '2',
hbase(main):040:1*   LIMIT => 3
hbase(main):041:1> }
ROW          COLUMN+CELL
 2          column=academic:faculty, timestamp=2026-05-23T13:56:25.321
           , value=medicine
 3          column=academic:faculty, timestamp=2026-05-23T13:56:33.793
           , value=engineering
 4          column=academic:faculty, timestamp=2026-05-23T13:56:43.637
           , value=arts
3 row(s)
Took 0.0142 seconds
```

Q4. Student 4 (nada) has moved from cairo to mansoura. Update her city.

Hint: put overwrites an existing value.

```
hbase(main):042:0> put 'student', '4', 'personal:city', 'mansoura'
Took 0.0143 seconds
```

Q5. Delete only the personal:age cell for student 3.

```
hbase(main):043:0> delete 'student', '3', 'personal:age'
Took 0.0080 seconds
```

Q6. Confirm the deletion worked by getting all data for row 3.

```
hbase(main):044:0> get 'student', '3'
COLUMN      CELL
academic:faculty timestamp=2026-05-23T13:56:33.793, value=engineering
academic:grade  timestamp=2026-05-23T13:57:33.667, value=A
academic:year   timestamp=2026-05-23T13:58:29.552, value=2
personal:city   timestamp=2026-05-23T13:54:26.754, value=giza
personal:name   timestamp=2026-05-23T13:52:47.518, value=omar
1 row(s)
Took 0.0151 seconds
```

Q7. Delete the entire row for student 5 (youssef).

```
hbase(main):045:0> deleteall 'student', '5'
Took 0.0100 seconds
```

Q8. Verify student 5 no longer exists with a full scan.

```
hbase(main):046:0> scan 'student'
ROW COLUMN+CELL
1 column=academic:faculty, timestamp=2026-05-23T13:56:14.238, value=engineering
1 column=academic:grade, timestamp=2026-05-23T13:57:06.087, value=A
1 column=academic:year, timestamp=2026-05-23T13:58:06.547, value=3
1 column=personal:age, timestamp=2026-05-23T13:55:02.007, value=21
1 column=personal:city, timestamp=2026-05-23T13:53:47.328, value=cairo
1 column=personal:name, timestamp=2026-05-23T13:50:10.879, value=ali
2 column=academic:faculty, timestamp=2026-05-23T13:56:25.321, value=medicine
2 column=academic:grade, timestamp=2026-05-23T13:57:22.832, value=B
2 column=academic:year, timestamp=2026-05-23T13:58:17.493, value=4
2 column=personal:age, timestamp=2026-05-23T13:55:16.665, value=22
2 column=personal:city, timestamp=2026-05-23T13:53:57.333, value=alex
```

```
File Edit View Search Terminal Help
3 column=academic:year, timestamp=2026-05-23T13:58:29.552, value=2
3 column=personal:city, timestamp=2026-05-23T13:54:26.754, value=giza
3 column=personal:name, timestamp=2026-05-23T13:52:47.518, value=omar
4 column=academic:faculty, timestamp=2026-05-23T13:56:43.637, value=arts
4 column=academic:grade, timestamp=2026-05-23T13:57:44.456, value=C
4 column=academic:year, timestamp=2026-05-23T13:58:40.610, value=4
4 column=personal:age, timestamp=2026-05-23T13:55:46.632, value=23
4 column=personal:city, timestamp=2026-05-23T14:06:10.642, value=mansoura
4 column=personal:name, timestamp=2026-05-23T13:52:59.348, value=nada
4 row(s)
Took 0.0667 seconds
```